A - A Unique Letter

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 100 points

Problem Statement

You are given a string S of length 3.
Print a character that occurs only once in S.
If there is no such character, print `-1` instead.

Constraints

- S is a string of length 3 consisting of lowercase English letters.

Input

Input is given from Standard Input in the following format:

$$S$$

Output

Print the answer. If multiple solutions exist, you may print any of them.

Sample Input 1

```
pop
```

Sample Output 1

```
o
```

`o` occurs only once in `pop`.

Sample Input 2

```
abc
```

Sample Output 2

```
a
```

`a`, `b`, and `c` occur once each in `abc`, so you may print any of them.

Sample Input 3

```
xxx
```

Sample Output 3

```
-1
```

No character occurs only once in `xxx`.

B - Better Students Are Needed!

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 200 points

Problem Statement

N examinees took an entrance exam.
The examinee numbered i scored A_i points in math and B_i points in English.

The admissions are determined as follows.

1. X examinees with the highest math scores are admitted.
2. Then, among the examinees who are not admitted yet, Y examinees with the highest English scores are admitted.
3. Then, among the examinees who are not admitted yet, Z examinees with the highest total scores in math and English are admitted.
4. Those examinees who are not admitted yet are rejected.

Here, in each of the steps 1. to 3., ties are broken by examinees' numbers: an examinee with the smaller examinee's number is prioritized. See also Sample Input and Output.

Print the examinees' numbers of the admitted examinees determined by the steps above in ascending order, separated by newlines.

Constraints

- All values in input are integers.
- $1 \le N \le 1000$
- $0 \le X, Y, Z \le N$
- $1 \le X + Y + Z \le N$
- $0 \le A_i, B_i \le 100$

Input

Input is given from Standard Input in the following format:

$$\begin{array}{llll} N & X & Y & Z \\ A_1 & A_2 & \ldots & A_N \\ B_1 & B_2 & \ldots & B_N \end{array}$$

Output

Print the examinees' number of the admitted examinees in ascending order, separated by newlines.

Sample Input 1

```
6 1 0 2
80 60 80 60 70 70
40 20 50 90 90 80
```

Sample Output 1

```
1
4
5
```

- First, 1 examinee with the highest math score is admitted.
  - Examinee 1 is tied with Examinee 3, scoring the highest 80 points in math, and the tie is broken by the examinees' numbers, so Examinee 1 is admitted.
- Then, among the examinees who are not admitted yet, 0 examinees with the highest English scores are admitted.
  - Obviously, it does not affect the admissions.
- Then, among the examinees who are not admitted yet, 2 examinees with the highest total scores in math and English are admitted.
  - First, among the examinees who are not admitted yet, Examinee 5 is admitted, scoring the highest total score of 160 points.
  - Next, among the examinees who are not admitted yet, Examinee 4 is tied with Examinee 6, scoring a total score of 150 points. The tie is broken by the examinees' numbers, and Examinee 4 is admitted.

Therefore, the examinees' numbers of the admitted examinees are 1, 4, and 5. Print them in ascending order.

Sample Input 2

```
5 2 1 2
0 100 0 100 0
0 0 100 100 0
```

Sample Output 2

```
1
2
3
4
5
```

All examinees may be admitted.

Sample Input 3

```
15 4 3 2
30 65 20 95 100 45 70 85 20 35 95 50 40 15 85
0 25 45 35 65 70 80 90 40 55 20 20 45 75 100
```

Sample Output 3

```
2
4
5
6
7
8
11
14
15
```

C - Changing Jewels

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 300 points

Problem Statement

Takahashi has a red jewel of level N. (He has no other jewels.)
Takahashi can do the following operations any number of times.

- Convert a red jewel of level n (n is at least 2) into "a red jewel of level $(n-1)$ and X blue jewels of level n".
- Convert a blue jewel of level n (n is at least 2) into "a red jewel of level $(n-1)$ and Y blue jewels of level $(n-1)$".

Takahashi wants as many blue jewels of level 1 as possible. At most, how many blue jewels of level 1 can he obtain by the operations?

Constraints

- $1 \le N \le 10$
- $1 \le X \le 5$
- $1 \le Y \le 5$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$N \quad X \quad Y$$

Output

Print the answer.

Sample Input 1

```
2 3 4
```

Sample Output 1

```
12
```

Takahashi can obtain 12 blue jewels of level 1 by the following conversions.

- First, he converts a red jewel of level 2 into a red jewel of level 1 and 3 blue jewels of level 2.
  - After this operation, Takahashi has 1 red jewel of level 1 and 3 blue jewels of level 2.
- Next, he repeats the following conversion 3 times: converting a blue jewel of level 2 into a red jewel of level 1 and 4 blue jewels of level 1.
  - After these operations, Takahashi has 4 red jewels of level 1 and 12 blue jewels of level 1.
- He cannot perform a conversion anymore.

He cannot obtain more than 12 blue jewels of level 1, so the answer is 12.

Sample Input 2

```
1 5 5
```

Sample Output 2

```
0
```

Takahashi may not be able to obtain a blue jewel of level 1.

Sample Input 3

```
10 5 5
```

Sample Output 3

```
3942349900
```

Note that the answer may not fit into a 32-bit integer type.

D - Draw Your Cards

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 400 points

Problem Statement

There is a deck consisting of N face-down cards with an integer from 1 through N written on them. The integer on the i-th card from the top is P_i.

Using this deck, you will perform N moves, each consisting of the following steps:

- Draw the topmost card from the deck. Let X be the integer written on it.
- Stack the drawn card, face up, onto the card with the smallest integer among the face-up topmost cards on the table with an integer greater than or equal to X written on them. If there is no such card on the table, put the drawn card on the table, face up, without stacking it onto any card.
- Then, if there is a pile consisting of K face-up cards on the table, eat all those cards. The eaten cards all disappear from the table.

For each card, find which of the N moves eats it. If the card is not eaten until the end, report that fact.

Constraints

- All values in input are integers.
- $1 \le K \le N \le 2 \times 10^5$
- P is a permutation of $(1, 2, \ldots, N)$ (i.e. a sequence obtained by rearranging $(1, 2, \ldots, N)$).

Input

Input is given from Standard Input in the following format:

$$\begin{array}{llll} N & K & & \\ P_1 & P_2 & \ldots & P_N \end{array}$$

Output

Print N lines.
The i-th line ($1 \le i \le N$) should describe the card with the integer i written on it. Specifically,

- if the card with i written on it is eaten in the x-th move, print x;
- if that card is not eaten in any move, print -1.

Sample Input 1

```
5 2
3 5 2 1 4
```

Sample Output 1

```
4
3
3
-1
4
```

In this input, $P = (3, 5, 2, 1, 4)$ and $K = 2$.

- In the 1-st move, the card with 3 written on it is put on the table, face up, without stacked onto any card.
- In the 2-nd move, the card with 5 written on it is put on the table, face up, without stacked onto any card.
- In the 3-rd move, the card with 2 written on it is stacked, face up, onto the card with 3 written on it.
  - Now there is a pile consisting of $K = 2$ face-up cards, on which 2 and 3 from the top are written, so these cards are eaten.
- In the 4-th move, the card with 1 written on it is stacked, face up, onto the card with 5 written on it.
  - Now there is a pile consisting of $K = 2$ face-up cards, on which 1 and 5 from the top are written, so these cards are eaten.
- In the 5-th move, the card with 4 written on it is put on the table, face up, without stacked onto any card.
- The card with 4 written on it was not eaten until the end.

Sample Input 2

```
5 1
1 2 3 4 5
```

Sample Output 2

```
1
2
3
4
5
```

If $K = 1$, every card is eaten immediately after put on the table within a single move.

Sample Input 3

```
15 3
3 14 15 9 2 6 5 13 1 7 10 11 8 12 4
```

Sample Output 3

```
9
9
9
15
15
6
-1
-1
6
-1
-1
-1
-1
6
15
```

E - At Least One

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 500 points

Problem Statement

You are given an integer M and N pairs of integers $(A_1, B_1), (A_2, B_2), \ldots, (A_N, B_N)$.
For all i, it holds that $1 \le A_i < B_i \le M$.

A sequence S is said to be a **good sequence** if the following conditions are satisfied:

- S is a contiguous subsequence of the sequence $(1, 2, 3, ..., M)$.
- For all i, S contains at least one of A_i and B_i.

Let $f(k)$ be the number of possible good sequences of length k.
Enumerate $f(1), f(2), \ldots, f(M)$.

Constraints

- $1 \le N \le 2 \times 10^5$
- $2 \le M \le 2 \times 10^5$
- $1 \le A_i < B_i \le M$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$\begin{array}{ll} N & M \\ A_1 & B_1 \\ A_2 & B_2 \\ \vdots & \\ A_N & B_N \end{array}$$

Output

Print the answers in the following format:

$$f(1) \quad f(2) \quad \ldots \quad f(M)$$

Sample Input 1

```
3 5
1 3
1 4
2 5
```

Sample Output 1

```
0 1 3 2 1
```

Here is the list of all possible good sequences.

- $(1, 2)$
- $(1, 2, 3)$
- $(2, 3, 4)$
- $(3, 4, 5)$
- $(1, 2, 3, 4)$
- $(2, 3, 4, 5)$
- $(1, 2, 3, 4, 5)$

Sample Input 2

```
1 2
1 2
```

Sample Output 2

```
2 1
```

Sample Input 3

```
5 9
1 5
1 7
5 6
5 8
2 6
```

Sample Output 3

```
0 0 1 2 4 4 3 2 1
```

F - Find 4-cycle

Time Limit: 2 sec / Memory Limit: 1024 MB

Score : 500 points

Problem Statement

We have a simple undirected graph G with $(S+T)$ vertices and M edges. The vertices are numbered 1 through $(S+T)$, and the edges are numbered 1 through M. Edge i connects Vertices u_i and v_i.
Here, vertex sets $V_1 = \{1, 2, \ldots, S\}$ and $V_2 = \{S+1, S+2, \ldots, S+T\}$ are both independent sets.

A cycle of length 4 is called a 4-cycle.
If G contains a 4-cycle, choose any of them and print the vertices in the cycle. You may print the vertices in any order.
If G does not contain a 4-cycle, print `-1`.

▶ What is an independent set?

Constraints

- $2 \le S \le 3 \times 10^5$
- $2 \le T \le 3000$
- $4 \le M \le \min(S \times T, 3 \times 10^5)$
- $1 \le u_i \le S$
- $S + 1 \le v_i \le S + T$
- If $i \neq j$, then $(u_i, v_i) \neq (u_j, v_j)$.
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$\begin{array}{lll} S & T & M \\ u_1 & v_1 & \\ u_2 & v_2 & \\ \vdots & & \\ u_M & v_M & \end{array}$$

Output

If G contains a 4-cycle, choose any of them, and print the indices of the four distinct vertices in the cycle. (The order of the vertices does not matter.)
If G does not contain a 4-cycle, print `-1`.

Sample Input 1

```
2 3 5
1 3
1 4
1 5
2 4
2 5
```

Sample Output 1

```
1 2 4 5
```

There are edges between Vertices 1 and 4, 4 and 2, 2 and 5, and 5 and 1, so Vertices $1, 2, 4$, and 5 form a 4-cycle. Thus, $1, 2, 4$, and 5 should be printed.
The vertices may be printed in any order. Besides the Sample Output, `2 5 1 4` is also considered correct, for example.

Sample Input 2

```
3 2 4
1 4
1 5
2 5
3 5
```

Sample Output 2

```
-1
```

Some inputs may give G without a 4-cycle.

Sample Input 3

```
4 5 9
3 5
1 8
3 7
1 9
4 6
2 7
4 8
1 7
2 9
```

Sample Output 3

```
1 7 2 9
```

G - Scalene Triangle Area

Time Limit: 3 sec / Memory Limit: 1024 MB

Score : 600 points

Problem Statement

We have an $N \times N$ grid. The square at the i-th row from the top and j-th column from the left in this grid is called (i, j).
Each square of the grid has at most one piece.
The state of the grid is given by N strings S_i:

- if the j-th character of S_i is `O`, then (i, j) has a piece on it;
- if the j-th character of S_i is `X`, then (i, j) has no piece on it.

You are given an integer M. Using this M, we define that a piece P placed at (s, t) covers a square (u, v) if all of the following conditions are satisfied:

- $s \le u \le N$
- $t \le v \le N$
- $(u - s) + \frac{(v-t)}{2} < M$

For each of Q squares (X_i, Y_i), find how many pieces cover the square.

Constraints

- N, M, X_i, Y_i, and Q are integers.
- $1 \le N \le 2000$
- $1 \le M \le 2 \times N$
- S_i consists of `O` and `X`.
- $1 \le Q \le 2 \times 10^5$
- $1 \le X_i, Y_i \le N$

Input

Input is given from Standard Input in the following format:

$$\begin{array}{ll} N & M \\ S_1 & \\ S_2 & \\ \vdots & \\ S_N & \\ Q & \\ X_1 & Y_1 \\ X_2 & Y_2 \\ \vdots & \\ X_Q & Y_Q \end{array}$$

Output

Print Q lines.
The i-th line ($1 \le i \le Q$) should contain the number of pieces that covers (X_i, Y_i) as an integer.

Sample Input 1